Vinay Somala

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Data Science | Machine Learning | Python | SQL | SAP S/4HANA MM | Supply Chain Analytics

SUMMARY

M.Sc. Applied Data Science & AI student at SRH University Hamburg, specialising in Supply Chain Analytics, with hands-on SAP S/4HANA MM training and a software engineering internship. I build end-to-end data pipelines, machine-learning models and KPI dashboards with Python and SQL, and back them with automated tests and data quality checks. My SAP training covers the full Procure-to-Pay process, so I understand the business processes behind purchasing, delivery and inventory data. Seeking a working-student role or internship in data science, data analytics, SAP MM or supply chain analytics in Hamburg or remote.

EDUCATION

SRH University Hamburg Oct 2025 – Sep 2027 (expected)
M.Sc. Applied Data Science & Artificial Intelligence *Hamburg, Germany*

- **Specialisation:** Supply Chain Analytics
- **Relevant coursework:** Machine Learning, Artificial Intelligence, Statistical Modelling, Database Systems, Demand Forecasting, Inventory Management

Vedavyasa Degree College, Sri Venkateswara University 2021 – 2024
B.Sc. Mathematics, Statistics & Computer Science *India*

- **Core subjects:** Mathematics, Statistics, Computer Science, Programming, Database Management

EXPERIENCE

Karthikeyan Software Solutions Pvt. Ltd. Jan 2023 – Jul 2023
Full Stack Engineer Intern *India*

- Built responsive web components with HTML, CSS, JavaScript and PHP for client web applications.
- Worked with MySQL for data storage and retrieval behind application features.
- Supported debugging, testing and Git-based code changes within the development team.

PROJECTS

Olist Retail Analytics – Sales, Delivery & Demand GitHub
Python, SQL, DuckDB, scikit-learn, Streamlit

- Built an end-to-end pipeline on 99,441 real e-commerce orders (2016–2018): raw CSVs loaded into DuckDB, cleaned in SQL staging and modelled as a star schema with 2 fact and 4 dimension tables.
- Defined and documented every KPI (revenue, average order value, delivery time, late rate, review score) and built monthly, category, region and seller KPI views.
- Wrote 20 automated data quality checks (duplicate keys, orphan foreign keys, impossible dates, revenue reconciliation) and pytest tests on a synthetic dataset, running in GitHub Actions on every push.
- Found that late orders average a 2.27 review score vs 4.29 for on-time orders; the late rate peaked at 19.0% in March 2018 and is 12.1% in Rio de Janeiro vs 4.5% in São Paulo.
- Forecast weekly demand for the 10 largest categories, compared gradient boosting with two baselines (best WAPE 22.6%) and turned the forecast error into safety stock and reorder points.

  Live dashboard: Open the live Streamlit dashboard

CareerTrack – Job Application Tracker GitHub
Python, Flask, SQLite, Docker, pytest

- Built a Flask web app to track job applications with search, status tracking, CSV import and export, and an analytics page.
- Covered the app with 22 automated tests, continuous integration and a Docker setup for reproducible deployment.

Cryptocurrency Price Direction Prediction GitHub
Python, scikit-learn, XGBoost, Pandas

- Engineered time-series features (returns, lagged values, moving averages, rolling volatility) to predict the next 15-minute price direction for five crypto pairs.

- Compared Logistic Regression, Random Forest and XGBoost on accuracy, F1, ROC-AUC and balanced accuracy with a time-based split (last 365 days held out) to avoid leakage; tuned XGBoost gave a 0.52% lift over the baseline.

Supermarket Sales Data Pipeline **GitHub**
Python, SQL, DuckDB, Pandas, Streamlit

- Built an ETL pipeline that merges 1,000 sales transactions with product costs, calculates revenue and profit, and loads analytics tables into DuckDB.
- Wrote SQL for KPIs (total sales, profit, sales by product line and city) and built a Streamlit dashboard with filters.

Demand Forecasting & Inventory Simulation
AnyLogistix, Supply Chain Analytics

- Configured customers, factories, distribution centres and demand inputs to model an end-to-end supply chain.
- Compared how alternative inventory and distribution decisions affect supply chain planning scenarios.

ODI Cricket Analytics Dashboard
Python, MongoDB Atlas, Streamlit, Plotly

- Built an interactive dashboard to explore and compare ODI statistics, with data organised in MongoDB Atlas and charts in Plotly.

SAP TRAINING

ERP4Students, Technische Hochschule Brandenburg Summer 2026 – Completed
SAP MM – Sourcing & Procurement with SAP S/4HANA *Online*

- Completed the Procure-to-Pay process in SAP Fiori and SAP GUI: purchase requisition, RFQ and quotation, purchase order, goods receipt and invoice verification.
- Created and maintained material and vendor master records, purchasing info records and source lists, and worked with the enterprise structure.

ERP4Students, Technische Hochschule Brandenburg Winter 2026/27 – In progress
SAP MM – Inventory Management & Invoice Verification *Online*

- Studying inventory management, goods movements, stock transfers and invoice verification in SAP S/4HANA.

SKILLS

Programming & Analysis: Python, SQL, Pandas, NumPy, Statistical Modelling, Exploratory Data Analysis, Excel (PivotTables, XLOOKUP)
Machine Learning: scikit-learn, XGBoost, Random Forest, Logistic Regression, Feature Engineering, Time-Series Forecasting, Model Evaluation
Data Engineering: ETL Pipelines, DuckDB, MySQL, SQLite, MongoDB Atlas, Data Quality Checks, pytest, GitHub Actions
BI & Visualisation: Power BI, Tableau, Streamlit, Plotly, Matplotlib, KPI Dashboards
SAP & Supply Chain: SAP S/4HANA MM, SAP Fiori, SAP GUI, Procure-to-Pay, Material and Vendor Master Data, Demand Forecasting, Safety Stock, Reorder Points, AnyLogistix
Tools: Git, GitHub, Docker, Flask, REST APIs, Jupyter, VS Code, Linux, AWS (basics)

CERTIFICATIONS

- **SAP:** Introducing Joule Studio, course completion badge (2026); SAP Joule Studio certification, exam scheduled for December 2026
- **AWS:** AWS Academy Graduate – Machine Learning Foundations (2022); AWS Academy Graduate – Cloud Foundations (2022)
- **DataCamp:** Data Analyst with Python track; SQL Certification track

LANGUAGES

English: Professional | **German:** A2 (learning) | **Telugu:** Native | **Hindi:** Professional